



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2014
Higher 1

BIOLOGY

8875/01

Paper 1 Multiple Choice

Friday, 19 September 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

- 1 Four students, **A**, **B**, **C** and **D**, who observed a root hair cell from a plant using an electron microscope, were asked to confirm the presence or absence within the cell of described cellular structures.

Which student made the correct set of observations?

	Structure with a double membrane inside which are stacks of flattened membranes	Area near the nucleus containing a pair of structures that are composed of microtubules	Structure with a double membrane with inner membrane infolded	Network of tubular-shaped membranous sacs with no ribosomes visible on outer surface of membranes
A	✓	✓	✓	✗
B	✓	✗	✗	✓
C	✗	✓	✓	✓
D	✗	✗	✓	✓

Key

✓ = present

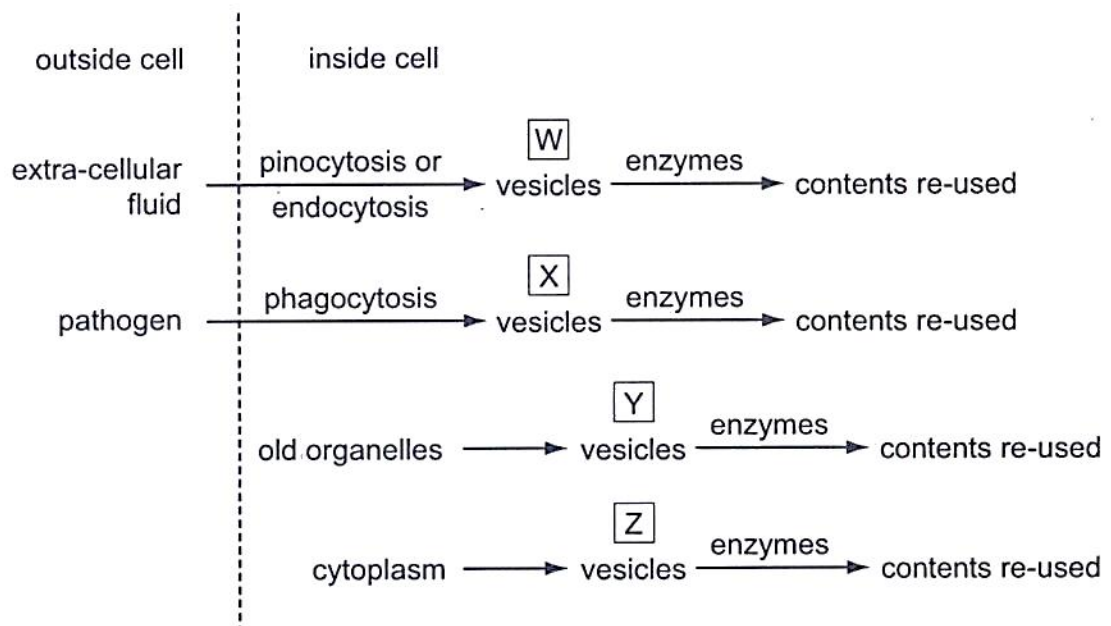
✗ = absent

- 2 A ribosome consists of a large and a small subunit, each subunit containing ribosomal RNA (rRNA) complexed with a protein.

Which sequence of events concerning ribosomes is correct?

- A** Within the nucleolus, rRNA and protein are synthesised and subunits formed. They become membrane bound as they are exported through the nuclear envelope to the cytoplasm and RER.
- B** rRNA and protein are synthesised in the Golgi body and are transported to the nucleolus for subunit formation.
- C** rRNA is synthesised in the SER and proteins are synthesised by the RER. Subunit formation occurs within the cytoplasm for free ribosomes and within the RER for attached ribosomes.
- D** rRNA synthesised within the nucleus is complexed with protein that has been imported from the cytoplasm. The subunits formed are exported to the cytoplasm via the nuclear pores.

- 3 The flow chart shows processes which takes place inside animal cells.



Which processes require the activity of lysosomes?

- A W, X, Y and Z
- B W and X only
- C X and Y only
- D Y and Z only

- 4 The table shows the results of an experiment on the rate of uptake of sugars by rat intestine.

sugars		rate of absorption relative to glucose	
		under aerobic conditions	under anaerobic conditions
hexose sugars	glucose	100	30
	galactose	106	32
pentose sugars	xylose	32	32
	arabinose	30	30

Which conclusion may be drawn from the results?

- A All four sugars can be absorbed by active transport.
 - B Only hexose sugars can be metabolised by the rat.
 - C The intestine is freely permeable to hexose sugars.
 - D The rate of passive diffusion of all four sugars into the gut is similar.
- 5 Four students, **A**, **B**, **C** and **D**, were given the same sequence of amino acids removed from a collagen molecule. Each student was asked to analyse the sequence and to explain how their analysis could be linked to a feature of collagen.

Which student's statement shows a correct feature of collagen linked to a correct analysis of the amino acid sequence?

- A Collagen has polypeptides arranged parallel to each other and the sequence contains a large variety of amino acids with different sized R-groups.
- B Collagen has polypeptides that are arranged very closely together and the sequence has every third amino acid as glycine.
- C Collagen has three polypeptides that can fold into a globular structure and the sequence contains cysteine and amino acids with hydrophobic R-groups.
- D Collagen is an insoluble molecule and the sequence contains a large proportion of amino acids with hydrophilic R-groups.

6 The key below can be used to separate the following carbohydrates:

amylopectin, cellulose, maltose, ribulose, sucrose

I	(a) readily soluble in water	-----	Go to II
	(b) insoluble in water	-----	Go to IV
II	(a) is a reducing sugar	-----	Go to III
	(b) is a non-reducing sugar	-----	V
III	(a) phosphorylated form is the carbon dioxide acceptor in photosynthesis	-----	W
	(b) product of starch digestion by amylase	-----	X
IV	(a) linked by $\alpha(1\rightarrow4)$ glycosidic bonds and $\alpha(1\rightarrow6)$ glycosidic bonds	-----	Y
	(b) linked by $\beta(1\rightarrow4)$ glycosidic bonds	-----	Z

	V	W	X	Y	Z
A	sucrose	maltose	sucrose	amylopectin	ribulose
B	maltose	ribulose	sucrose	cellulose	amylopectin
C	cellulose	ribulose	maltose	sucrose	amylopectin
D	sucrose	ribulose	maltose	amylopectin	cellulose

- 7 Phospholipids, glycolipids, glycoproteins and proteins are all molecules found in the membranes of cells.

Which type of bond is present in each type of molecule?

Bond types: 1 ester
 2 glycosidic
 3 ionic
 4 peptide

	Phospholipid	Glycolipid	Glycoprotein	Protein
A	1	2	4	3
B	1	3	2	4
C	3	4	1	2
D	3	1	2	4

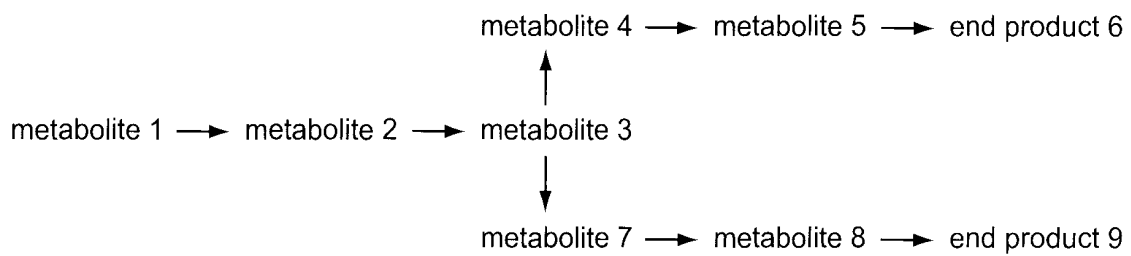
- 8 The enzyme rennin is found in gastric juice of young mammals. It causes the clotting of milk protein. The activity of rennin was investigated by recording the time taken for rennin to clot milk in different conditions. The table shows the different conditions used and the results of the investigation.

Tube number	solutions added to 10cm ³ of milk at 35°C						time for milk to clot/min
	2% renin	water	calcium nitrate	calcium citrate	lead nitrate	hydrochloric acid	
1	5 cm ³		2 cm ³			2 cm ³	1
2	5 cm ³				2 cm ³	2 cm ³	no clot formed
3		5 cm ³	2 cm ³			2 cm ³	no clot formed
4	5 cm ³	2 cm ³		2 cm ³			25
5	5 cm ³	2 cm ³				2 cm ³	10
6	5 cm ³			2 cm ³		2 cm ³	2

What is a correct conclusion?

- A** Calcium ions increase the activity of rennin.
- B** Citrate ions are necessary for the activity of rennin.
- C** Hydrochloric acid is necessary for the activity of rennin.
- D** Nitrate ions inhibit the activity of rennin.

- 9 The diagram shows a metabolic pathway controlled by end product inhibition.



An increase in the concentration of the end product increases above a set level, an enzyme in the pathway leading to the end product is inhibited.

An increase in the concentration of end product **6** does not lead to a decrease in the synthesis of end product **9**.

Which enzyme is inhibited by end product **6**?

- A metabolite 1 → metabolite 2
 - B metabolite 2 → metabolite 3
 - C metabolite 3 → metabolite 4
 - D metabolite 3 → metabolite 7
- 10 A chemical known to affect mitosis was added, at different stages of mitosis, to actively dividing plant cells with 12 chromosomes.

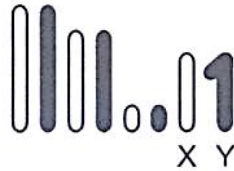
The results showed that adding this chemical during prophase resulted in cells with 24 chromosomes. Adding the chemical at any other stage resulted in cells with 12 chromosomes.

Which process during mitosis is affected by this chemical?

- A condensing of chromosomes
- B organizing the spindle
- C producing the centromeres
- D separating centrioles

- 11** No crossing over occurs during meiosis in male fruit flies of the species *Drosophila melanogaster*.

The diagram shows the four pairs of homologous chromosomes present in a testis cell of a male fly.

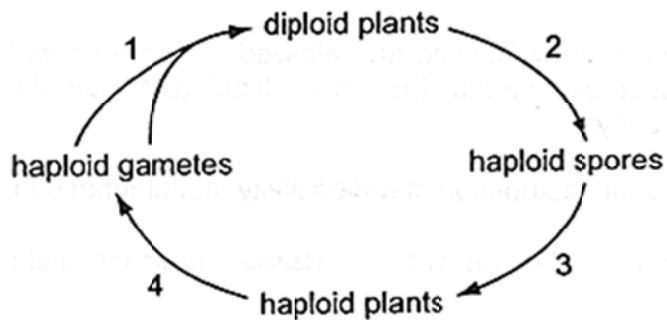


Which set of chromosomes in a gamete nucleus shows the genetic variation resulting from independent assortment?



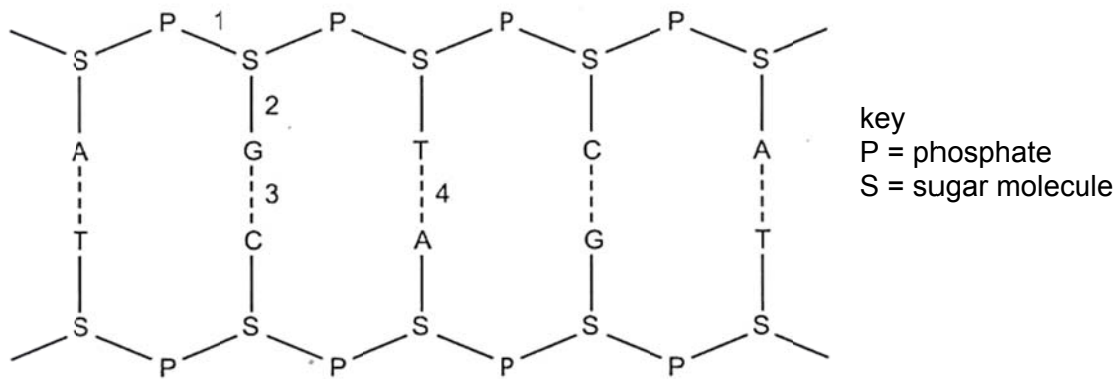
- 12** The diagram shows the life cycle of a type of plant.

When does meiosis occur?



- A** 1 and 3 **B** 2 only **C** 2 and 4 **D** 4 only

13 The diagram shows part of a nucleic acid.



Which row correctly describes the bonds shown in the diagram at positions 1, 2, 3 and 4?

	Is formed by condensation	Forms a di-ester	Occurs during transcription	Involves attraction between polar molecules
A	1	2	1,2 and 3	3
B	1 and 2	1	3 and 4	3 and 4
C	2,3 and 4	1	3 and 4	1,3 and 4
D	2	3	1,3 and 4	4

14 The structure of a DNA molecule has the features shown below.

- 1 Four different bases
- 2 Antiparallel strands
- 3 Pentose sugars
- 4 Phosphate groups
- 5 Supercoiling

Which features make DNA suitable for storing information?

- A** 1, 2 and 5
- B** 1, 4 and 5
- C** 2, 3 and 4
- D** 3, 4 and 5

- 15** The active messenger RNAs (active mRNAs) in tissue cells can be isolated by passing the homogenized cell contents through a fractionating column. The column has short lengths of uracil nucleotides attached to a solid supporting material. Most molecules of mRNA that pass through the column quickly break up into small pieces and cannot be translated.

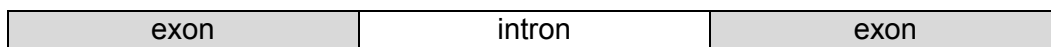
The active mRNAs that attach to the column can be separated again by appropriate treatment.

Which statements correctly describe active mRNAs?

- 1 Active mRNAs can be held to the fractionating column by bonds between adenine and uracil bases.
- 2 Active mRNAs can be released from the fractionating column by breaking hydrogen bonds.
- 3 Only mRNAs with polyadenine tailing can be translated.
- 4 Polyadenine tailing stabilizes mRNA and prevents it from being broken up.

- A** 1 and 2
- B** 1 and 3
- C** 2 and 3
- D** 1, 2 and 3

- 16** The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

- A** activation of the gene by a steroid hormone
- B** attachment of mRNA to a ribosome
- C** binding of RNA polymerase to DNA
- D** splicing of pre-mRNA

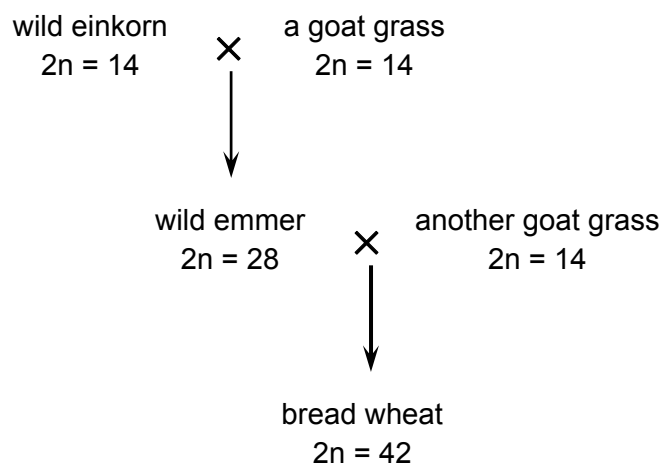
17 The processing of pre-mRNA involves the following steps to produce functional mRNA.

- 1 polyadenylation
- 2 removal of introns
- 3 splicing of exons
- 4 capping

In which order do they occur?

- A** 1 → 3 → 2 → 4
- B** 2 → 4 → 1 → 3
- C** 3 → 1 → 4 → 2
- D** 4 → 2 → 3 → 1

18 Bread wheat is the result of crosses between ancient wheat species and goat grasses which have occurred in several stages over the last ten thousand years.



Why does this type of chromosome change **not** normally happen?

- A** Diploid cells are formed by mitosis.
- B** Diploid gametes undergo a reduction division.
- C** Meiosis halves the diploid number of chromosomes.
- D** Meiosis results in diploid gametes.

- 19** The speech defect known as stuttering may involve two genes, G and N. Most people who are homozygous for the alleles g and n are not stutterers.

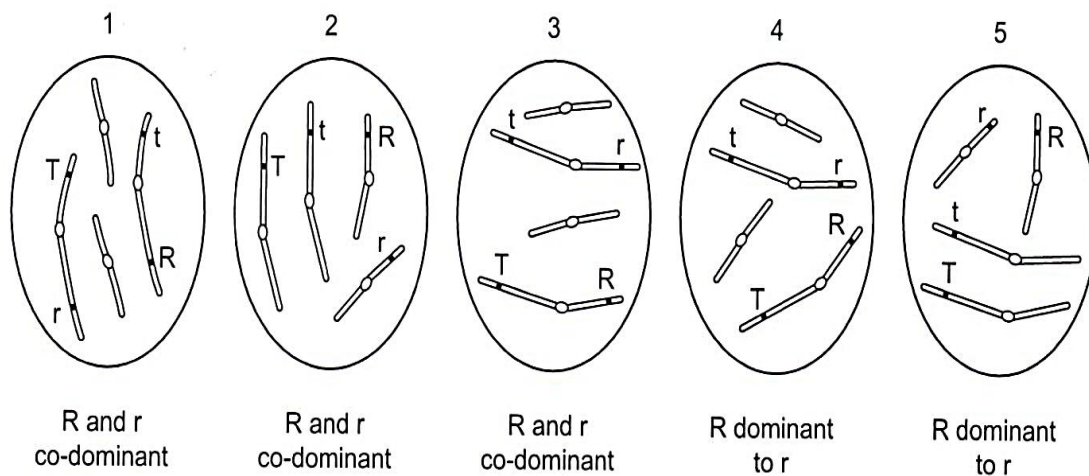
However, recent research has shown that the presence of either of the mutant alleles G or N can cause stuttering in heterozygotes.

Using this information, which proportion of the children of a couple, the father with genotype Ggnn and the mother ggNn, are likely to be stutterers?

- A** 3/16 **B** 6/16 **C** 9/16 **D** 12/16

- 20** In a family of flowering plants, height is controlled by a pair of alleles. The allele for tall (T) is always dominant to the allele for short (t). The flower colour is also controlled by a pair of alleles. In some species, the allele for red (R) is dominant to the allele for white (r). In other species the colour alleles are co-dominant. (For simplicity, the symbols R and r are used for the co-dominant alleles.)

The diagram shows the chromosome arrangement and information about the height alleles and the flower colour allele in five species of this family of plants.



Each of the plants 1, 2, 3, 4 and 5 was test crossed.

Assuming there is no crossing over, which plants would produce offspring with the phenotypes short with white flowers and tall with red flowers?

- A** 1, 3 and 4
B 2, 4 and 5
C 2 and 5 only
D 4 and 5 only

- 21 The incidence of primaquine sensitivity in a population is 14% in men and 2% in women. A proportion of the remaining women, but not the men, exhibited mild sensitivity.

What does this suggest about the inheritance of this defect?

- A autosomal, codominant
- B autosomal, recessive
- C sex-linked, codominant
- D sex-linked, recessive

- 22 Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?

- A Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.
- B Identical genetic crosses performed between varieties of *Dendranthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glasshouse.
- C The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parent plants.
- D The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day length, of a height varying between 55.0 – 71.0 cm.

23 Which of the following statements are true for both cyclic and non-cyclic phosphorylation?

- I Hydrolysis of ATP occurs in both cyclic and non-cyclic phosphorylation.
- II Electron displacement occurs in the reaction centre of photosystem I in both cyclic and non-cyclic phosphorylation.
- III Energy released from the electron transport chain is used to pump protons from the stroma into the thylakoid lumen.
- IV NADP^+ is oxidized in both cyclic phosphorylation and non-cyclic phosphorylation.

- A I and IV
- B II and III
- C II and IV
- D II, III and IV

24 Which of the following processes do **not** occur during the conversion of glucose to two molecules of pyruvate?

- I hydrolysis of ATP
- II phosphorylation of hexose
- III release of CO_2
- VI reduction of NAD
- V oxidative phosphorylation of ADP

- A I and III only
- B II and IV only
- C III and V only
- D I, II and V only

- 25** Many types of evidence, including homology, can provide evidence in support of Darwin's theory of natural selection.

Which statement does **not** provide support?

- A** The allele for sickle cell haemoglobin that gives resistance to malaria is more frequent in malarial areas.
- B** The distribution of variants of the A blood group antigen reflects human migration patterns.
- C** The homozygous condition of the sex-linked allele for a non-functional blood clotting protein is rare.
- D** The molecular structure of ATP is almost identical in all eukaryotes.

- 26** The pigment haemoglobin found in red blood cells of mammals and birds combines readily with oxygen.

DNA analysis has revealed that a form of haemoglobin is found in a wide range of unrelated phylogenetic groups including bacteria, annelids, arthropods and leguminous plants.

Which evolutionary processes could account for the distribution of haemoglobin in such a wide variety of organisms?

	Adaptive radiation	Conservation of genes	Natural selection
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

Key

✓ = present

✗ = absent

- 27 A large population of equal numbers of dark and light mice was released into an area where owls are predators of mice. Because of predation, only 25% of these mice survived and selective killing by owls changed the proportions of dark to light mice from 1: 1 to 4: 1.

If mice produce an average of eight offspring per litter and the pattern of predation remains the same, what would happen to the population of light mice?

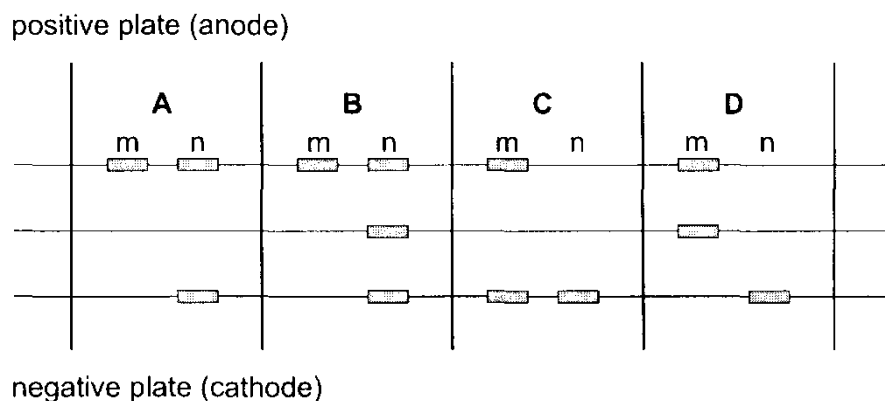
- A They disappear after one further generation.
- B They disappear after two further generations.
- C They remain at the level of one fifth of the population.
- D They will be reduced to a constant but very low frequency.

- 28 There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a wrinkled skin and the other with a smooth skin.

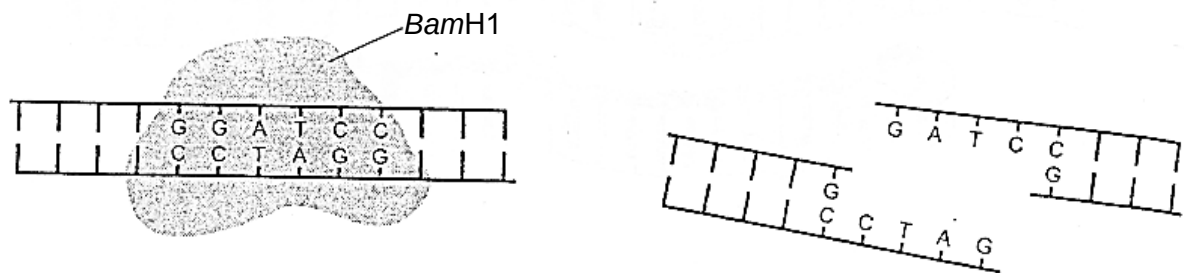
Plants with wrinkled seeds have a recessive mutation caused by the insertion of a transposable element into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the DNA for this gene removed and separated by electrophoresis.

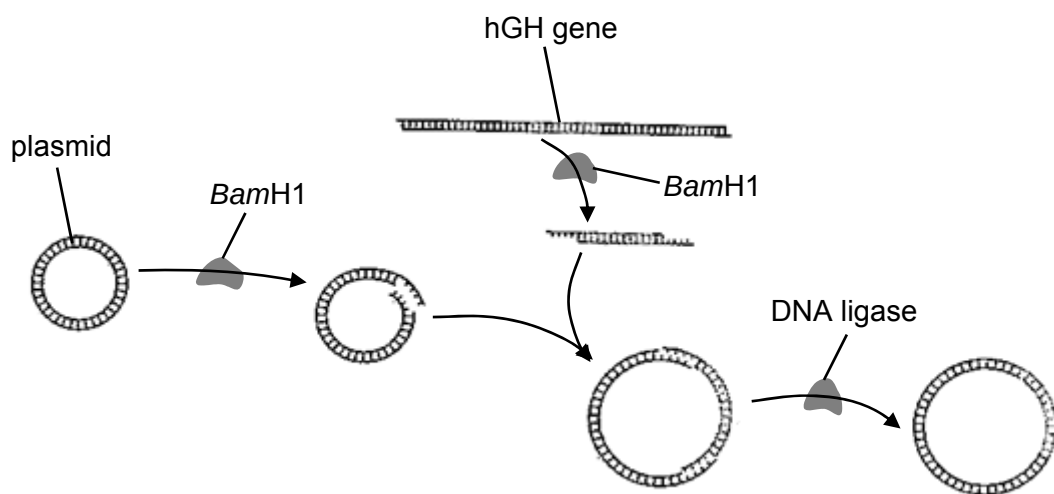
Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?



- 29 *Bam*H1 is a restriction enzyme that cuts DNA as shown in the diagram.



The diagram below shows part of the procedure for producing *E.coli* that will synthesise human growth hormone, hGH.



At the end of this process, many plasmids do not contain the hGH gene.

What could explain this?

- A Different alleles of the hGH gene have different sticky ends.
- B Not all of the plasmids cut by *Bam*H1 have sticky ends.
- C Some of the plasmids are cut at more than one position.
- D The sticky ends of some of the plasmids rejoin with each other.

- 30** Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

- A** salmon providing eggs for treatment
- B** trout providing reproductive organ tissue
- C** young triploid salmon
- D** young trout